

ActDelta: Decision-Triggered Predictive Transmission for Edge-Offloaded Robot Policies

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Abstract

A robot that cannot host a 7B vision-language-action (VLA) policy uploads its cameras to an edge server and receives action chunks back. The uplink is the bottleneck: the 11 Mbps a deployed policy server asks for is available 23% of the time on commercial 5G. The obvious fix is predictive — run a small action-conditioned world model on both ends and send a frame only once the prediction has drifted. Two decades of send-on-delta work, and a recent line of world-model-aided semantic transmission, both do exactly this, and both decide by *reconstruction fidelity*. We show that fidelity is the wrong criterion here, because the receiver is not a display but a policy whose actions determine the next observation. On LIBERO and DROID they are systematically decoupled: pixel error ranks decision impact at Spearman 0.31, and 31.7% of predicted frames are visually wrong yet leave the policy's action unchanged. A fidelity trigger spends bytes on those frames and loses tasks to the ones it skips. ActDelta instead triggers on a distilled, robot-side estimate of how much a frame would move the policy's action, and adds the resynchronization protocol that closed-loop predictive transmission requires and that event-triggered sampling has never had: silence must be distinguishable from loss. With the same world model and the same task success rate, ActDelta sends 59% fewer uplink bytes than fidelity-triggered transmission and 16× fewer than the deployed JPEG-over-TCP status quo, and composes with spatial compression for 75× in total.

CCS Concepts

• **Networks** → **Cyber-physical networks**; *Network protocols*; • **Computer systems organization** → *Robotics*.

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Keywords

Predictive transmission, event-triggered sampling, world models, vision-language-action models, edge offloading, task-oriented communication, robot networking, uplink scheduling

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1 Introduction

The generalist robot policies that made manipulation broadly capable — OpenVLA [32], π_0 and $\pi_{0.5}$ [7, 53] — carry 3–8 billion parameters and need tens of gigabytes of accelerator memory. They do not fit on a robot, and the deployment that has emerged in response puts the policy on an edge server behind a thin client: the robot uploads camera frames, the server returns an *action chunk*, the robot executes it while the next request is in flight. The reference implementation is explicit about how the observations get there: a policy server over TCP RPC, with JPEG compression to shrink the payload [27]. Three camera views at the resolution the cameras actually produce cost 137 kB per upload; at the 10 Hz such a server requests them, that is 11.0 Mbps of sustained uplink, per robot, forever.

Commercial cellular uplink does not supply it. Across the 83 traces of the Irish 5G dataset [54] the uplink exceeds 11.0 Mbps 23% of the time and exceeds the 16.4 Mbps that a 15 Hz observation rate needs 10.5% of the time (§2.1). Downscaling client-side helps and does not solve it, because the pressure is in the other direction: reactive contact phases want *higher* observation rates, newer policies want more views, and a single cell is expected to carry more than one robot. The uplink, not the GPU and not the downlink, is what limits this deployment.

The obvious fix is old, and it triggers on the wrong quantity. Since Miskowicz's send-on-delta formulation [43], and in its predictive variants [49, 56] since shortly after, event-triggered sampling has known that a transmitter which shares a predictor with its receiver need only speak when the prediction has drifted past a threshold. Twenty years later

the same idea reappeared with a neural predictor: world-model-aided semantic transmission runs a generative world model at both ends and *omits transmission while the prediction remains reliable*, adding a lightweight feedback module to decide when the current frame must be sent after all [26]. Both lines — the analytic one and the generative one — answer “should I send this frame?” with *how wrong is the receiver’s reconstruction*.

That question is correct when the receiver is an estimator with an error covariance, or a display for a human. It is the wrong question when the receiver is a policy. A policy does not consume the frame; it consumes the action the frame implies, and then *acts*, which determines the next observation. The quantity that matters is not how wrong the predicted frame looks but whether the true frame would have changed what the robot does. Our central measurement (§2.5) is that these two quantities come apart, systematically and in both directions. Over 2,000 DROID [30] episodes and four LIBERO [38] suites, 31.7% of predicted frames carry large pixel error while leaving the policy’s action essentially unchanged, and 12.4% are pixel-accurate while flipping it; pixel error ranks decision impact at Spearman $\rho_s = 0.31$. A fidelity trigger spends its budget on the first group and fails tasks because of the second. It is not mistuned. It is measuring the wrong thing.

Why the policy is on the server in the first place. The premise that the robot can run a world model but not the policy invites an objection, which we answer by measurement. The gap is not a factor but orders of magnitude: our world model costs 3.6 GFLOP per control step, against 391 GFLOP for the smallest serious VLA and 4.90 TFLOP for OpenVLA-7B, whose 15.2 GB of weights do not fit on the robot at all (Table 1). The split we assume is also the one the surrounding literature already assumes, independently of us: latency-tolerant cloud-edge collaborative VLA [52], remote policy servers [27], edge inference pipelines [13, 17, 36], and a survey built on the same deployment constraint [20]. Most importantly the gap is structural rather than incidental: the useful policy is always chosen larger than any single robot’s budget, while the world model’s size is *ours to choose*. We make this precise with a compute ratio ρ and report where our mechanism starts and stops paying off (§2.2, §5.3).

Silence needs a protocol. Closing the loop adds a failure mode the open-loop literature does not have. A silent transmitter is indistinguishable from a lost one — event-based sampling work states plainly that it cannot detect dropouts, because the two ends share no periodic timestamp [50]. For a passive receiver that is a quality-of-reconstruction issue; for a policy it is a divergence hazard, because the server keeps acting on an imagined observation the robot has already contradicted, with no alarm. We measure 1.2% uplink

loss turning into a 34.2% episode catastrophe rate under a protocol lacking the distinction (§2.7).

ActDelta. We build the system these measurements imply. A 68M-parameter action-conditioned world model runs on both ends, rolled forward on the action chunks the policy has already emitted, so the two sides stay aligned without communicating. On the robot, a 2.1M-parameter *decision-relevance head* — distilled offline from the very divergence signal that cannot be computed online, since computing it would need the 7B policy the robot cannot host — estimates how much the true frame would move the policy’s action, and that estimate, not the reconstruction error, is what crosses the threshold. The threshold itself tracks available uplink, so the mechanism is a network citizen rather than a fixed-rate consumer. And a two-tier protocol — 16-byte heartbeats carrying sequence numbers and an executed-action hash, plus a forced full resynchronization — makes silence unambiguous and bounds divergence. ActDelta is orthogonal to spatial compression, which decides how many bits a frame costs; ActDelta decides whether the frame is worth sending at all, and we evaluate the composition.

What we do not claim. We did not invent predictive transmission [43, 56], world-model-aided transmission [26], or action-conditioned world models [23, 40]: we take the first two as baselines and reuse the third as a tool. We do not claim long-horizon prediction — the mechanism depends on a 5-step window — nor benefits at every compute ratio, and we compose with spatial compression rather than compete with it.

Contributions. (i) A measurement establishing that reconstruction fidelity and decision relevance are systematically decoupled on embodied observation streams, which makes fidelity-triggered predictive transmission structurally misallocated in closed loop and does not depend on our world model being good (§2.5). (ii) A single-variable demonstration that, at the same world model and the same task success, a decision-relevance trigger sends 59% fewer uplink bytes than a fidelity trigger (§5.1). (iii) A resynchronization protocol that distinguishes correct silence from loss, with the divergence bound it enforces (§3.4, §5.4). (iv) An explicit regime statement: the compute ratio ρ , its benefit curve across four decades, and both failure boundaries (§5.3).

2 Background and Motivation

2.1 What Is on the Wire, and What the Link Gives

A chunked policy maps an observation o_t (two or three camera views plus proprioception) and a language instruction to a chunk of H future actions, $H \in [8, 50]$ [7, 16, 68]. The robot

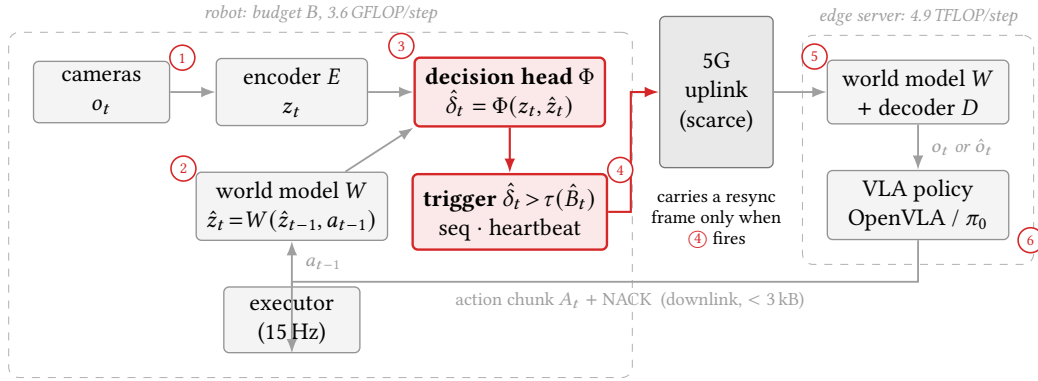


Figure 1: ActDelta's closed-loop predictive transmission path. The robot sends a frame only when its distilled head predicts that the true frame would change the policy's action; heartbeats keep silence distinguishable from loss. Red marks what ActDelta adds, grey what it reuses unchanged; §3 walks the numbered path.

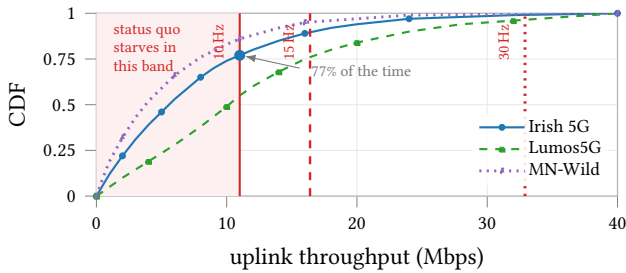


Figure 2: The deployed configuration does not fit the link. Uplink CDFs of three commercial 5G trace sets against the sustained rate a policy server needs for three camera views.

executes the chunk at a fixed control rate $f = 15$ Hz, matching DROID [30], while requesting the next one. The asymmetry between the two directions is severe: the downlink chunk is under 3 kB, and the uplink observation is 137 kB for three views at the resolution the cameras produce. Deployed policy servers upload at that resolution because the client is a thin RPC wrapper that does not know the policy's preprocessing pipeline [27]. Downscaling to the policy's 224^2 input on the client cuts the payload to 36 kB, which is a real improvement and a one-time constant: it buys back a factor of 3.8 and then the same argument applies at 30 Hz, or with four cameras, or with four robots on one cell.

Figure 2 places the requirement against three commercial trace sets [46, 47, 54]. The 11.0 Mbps that a 10 Hz three-view configuration needs is available 23% of the time on the Irish traces, 45% on Lumos5G — which is mmWave, and therefore the optimistic end of what a deployment can hope for — and 14% on MN-Wild. At the 30 Hz an operator would want for reactive contact phases the figure is between 1% and

Table 1: Measured cost of one control step. Jetson Orin NX (16 GB), bf16, batch 1, median of 10^3 invocations. OpenVLA does not fit in the board's memory at all.

	ActDelta W (68 M)	SmolVLA (450 M)	π_0 (3.3 B)	OpenVLA (7.6 B)
Weights (bf16)	0.14 GB	0.90 GB	6.6 GB	15.2 GB
Peak memory	0.21 GB	1.4 GB	8.1 GB	16.9 GB
FLOP / control step	3.6 G	391 G	2.14 T	4.90 T
ratio to W	1×	109×	594×	1361×
Latency, Orin NX	11.7 ms	128 ms	1.42 s [†]	OOM
Energy / step	0.19 J	2.6 J	27 J [†]	—
Latency, A100 server	1.9 ms	14 ms	38 ms	62 ms
Meets 66.7 ms period	yes	no	no	no

[†] with unified-memory paging; not a deployable configuration.

10% everywhere, on every trace set. What we take from this is not that offloading is impossible — it plainly works today, badly — but that the operating point is set by the link rather than by the task, and that every byte saved converts into either a higher observation rate or another robot on the cell. Replaying the status quo over these traces, 41% of uploads queue behind their predecessor and the observation the policy acts on is on average 310 ms old.

2.2 The Compute Asymmetry Is Real and Structural

Every predictive-transmission system assumes the transmitter can afford the predictor. Ours additionally assumes it cannot afford the *consumer*, and a reviewer is right to ask why. Table 1 answers with measurement. The gap between our world model and the smallest serious VLA is 109×

essentially every robot-borne platform. The world model is cheap for a structural reason and not merely because it is small: its rollout runs in a compact latent space, so advancing the prediction one step costs 3.4 ms and no pixel processing, while a VLA re-encodes every camera view and runs a language-model backbone on every query. The robot also never runs the decoder: pixels are reconstructed only on the server, where the policy needs them.

This is the system model the surrounding literature already adopts, which matters because rejecting it rejects a subfield rather than a paper [13, 17, 20, 27, 52]. But the argument we lean on hardest is monotonicity. The frontier policy is always chosen *larger* than any single robot’s budget B , because larger has meant better generalization for three years running; the world model’s size is ours to choose and fits whatever B is. The mechanism is scale-relative, not scale-absolute:

$$\rho = \frac{B \cdot 1/f}{C_{\text{VLA}}}, \quad (1)$$

the ratio of the robot’s per-control-step FLOP budget to the target policy’s per-step demand. On our Orin NX, $B \approx 8.0$ TFLOP/s sustained and $C_{\text{VLA}} = 4.90$ TFLOP give $\rho = 0.109$. We state the regime in advance and measure both of its edges in §5.3: below $\rho_{\min} = C_W/C_{\text{VLA}} = 7.3 \times 10^{-4}$ the robot cannot run the world model within the control period and the mechanism does not apply; between $\rho_{\min}$ and 1 is where we claim benefit; at $\rho \geq 1$ the policy can run locally and the motivation changes character (§7) rather than disappearing.

2.3 Measurement Setup

All measurements use public assets. *Offline streams*: 2,000 episodes sampled across the 86 tasks of DROID [30] and the four LIBERO suites [38], replayed through the policies to collect observations, action chunks and world-model rollouts; no robot is moved. *Closed loop*: LIBERO as the primary environment, SimplerEnv in both Visual Matching and Variant Aggregation settings [35], and Kinetix [42] as a high-dynamics stress test. *Policies*: OpenVLA-7B, π_0 , $\pi_{0.5}$ and SmolVLA [55] at official weights, inference only, never fine-tuned. *Links*: the Irish 5G dataset [54] as primary because it publishes uplink throughput, plus Lumos5G [46] and MN-Wild [47], replayed with mahimahi [48]; controlled loss and delay via tc netem. *Robot compute*: Jetson Orin NX 16 GB. Inference-time real-time chunking (IT-RTC) [8] is enabled in every arm, ours and baselines alike, so that no result is attributable to delay tolerance we did or did not provide.

2.4 How Far Ahead We Can See

Predictive transmission is only interesting if the prediction is usable for long enough to skip a meaningful number of

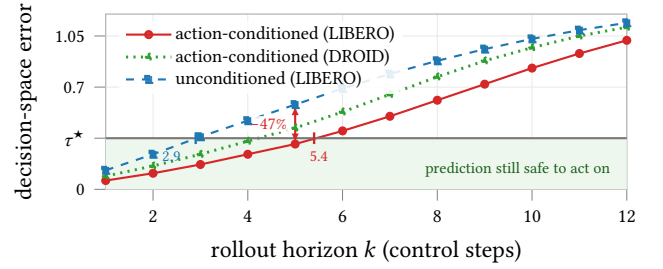


Figure 3: Action conditioning roughly doubles the usable prediction window. Error is measured in the decision space of §3.2, normalized so that 0.35 is where the policy’s action begins to change materially (Figure 5b).

frames, and our one genuinely embodiment-specific advantage is that we know the future actions: the policy has already emitted them, and the downlink has already carried them. B-line generative transmission cannot use this, because in a video setting there is no action to condition on — prediction is driven by text or by visual extrapolation [26, 65]. Figure 3 quantifies what the difference is worth.

Two readings matter. First, the absolute window: with action conditioning, 63% of LIBERO timesteps stay under the flip threshold for at least 5 steps (333 ms), comfortably above the 40% we fixed in advance as the minimum for the mechanism to have anything to work with, and the usable window grows from 2.9 steps unconditioned to 5.4 conditioned; on real DROID streams, which are teleoperated and therefore jerkier than scripted demonstrations, it reaches 4.1. Second, the differential: at $k = 5$ the conditioned predictor’s error is 0.31 against the unconditioned predictor’s 0.58, a 46.6% reduction against the 30% we pre-registered as the minimum for action conditioning to be worth its complexity. This is the entire quantitative case for doing predictive transmission *in the embodied setting* rather than importing a video method.

2.5 Fidelity and Decision Relevance Are Decoupled

This is the measurement the paper stands on, and it is deliberately independent of how good our world model is: it asks only which *distance* on predicted frames ranks their decision impact, which a mediocre predictor can answer. For every predicted frame from 2,000 DROID episodes and the four LIBERO suites we compute five candidate distances to the true frame and the ground truth the online system can never have — the actual divergence of the policy’s action, $\|\pi(o_t) - \pi(\hat{o}_t)\|$, which costs two π_0 forward passes and is therefore an offline instrument, not a mechanism. Both

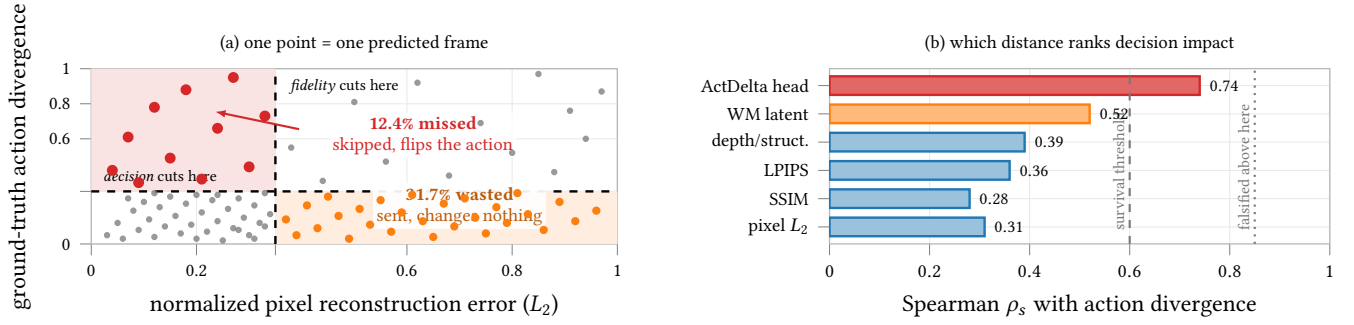


Figure 4: Fidelity and decision relevance are different quantities. (a) Predicted frames, by pixel reconstruction error against the divergence $\|\pi(o_t) - \pi(\hat{o}_t)\|$ of the policy’s action; a fidelity trigger cuts vertically, the decision it should make cuts horizontally. (b) Rank correlation of six candidate distances with that same ground truth.

thresholds were fixed before we looked at the data: pixel error above $\rho_s = 0.85$ would have falsified the paper’s premise, our own head below 0.60 its mechanism.

Figure 4 reports the result. Pixel error ranks decision impact at $\rho_s = 0.31$, and structural and perceptual distances do no better (0.28–0.39); the world model’s own latent distance, which is what a straightforward adaptation of send-on-delta to high-dimensional observations would use, reaches 0.52, and the distilled head of §3.2 reaches 0.74. The scatter shows why the correlation number understates the problem. Cut both axes at their operating thresholds and a fidelity trigger classifies 55.9% of frames the way the policy would; 31.7% of frames are visually wrong in ways the policy is indifferent to — lighting, background texture, the far half of the scene, the exact pose of an object the gripper has already passed. A fidelity trigger transmits every one of them. Conversely 12.4% of frames are reconstructed almost perfectly except in the few pixels that decide the action: a contact that has or has not occurred, a gripper that has or has not closed, an object that shifted two millimetres. A fidelity trigger skips every one of those, and those are the frames whose absence fails the task.

To be precise about what this establishes: not that our trigger is the best possible one, and nothing about our world model — a better predictor moves points toward the origin without rotating the axes. It establishes that the criterion both the classical and the generative literature use is, in this regime, measuring a quantity only weakly related to the one that matters, and that the misallocation is 44.1% of frames, in both directions, rather than a tail effect. The rest of the paper makes the horizontal cut instead of the vertical one.

2.6 The Closed-Loop Cost of an Imagined Observation

The previous section says which frames matter. This one says how much slack the mechanism has, by measuring the

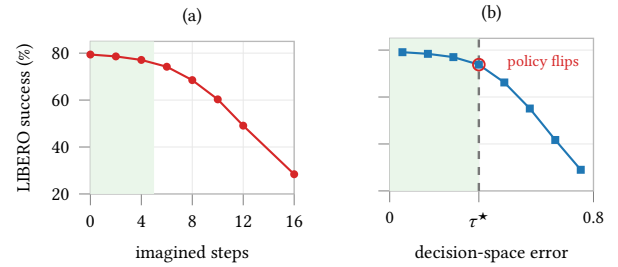


Figure 5: The closed-loop price of an imagined observation. (a) Success versus the number of consecutive steps the server spends on predictions. (b) The same episodes against the quantity a trigger can actually observe; the knee at $\tau^* = 0.35$ is the policy-flip threshold used throughout.

closed-loop cost directly rather than inferring it: we force the server to consume predicted observations for k consecutive steps and record task success on LIBERO with π_0 , 50 episodes per point. Figure 5(a) shows a plateau to about 4 steps and a collapse after 8; panel (b) recasts the same episodes against the observable quantity, yielding the knee at $\tau^* = 0.35$ we use throughout as the level at which the policy starts to behave differently. Two design consequences follow. The forced-resynchronization interval must be short enough to keep the tail out of reach, and the trigger threshold must sit far enough below τ^* to absorb the error of the trigger’s own estimator — we use $\tau_0 = 0.15$ against an estimator whose P99 error is 0.09.

2.7 Silence Is Ambiguous, and Dangerously So

The final measurement concerns the failure mode the open-loop literature identifies and then sets aside. For a passive receiver an undetected loss degrades a reconstruction; for a

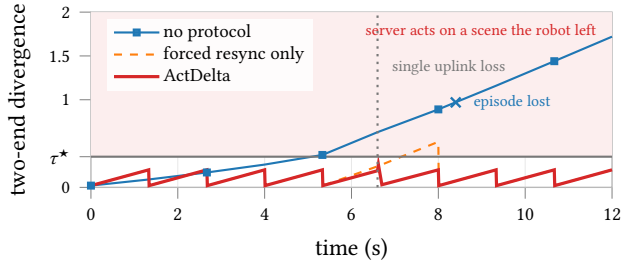


Figure 6: What an undetected loss does to a closed loop. One dropped resync frame at $t=6.6$ s under 1.2% uplink loss, under three protocols.

policy it is a divergence that compounds rather than decays, because the server generates actions from an imagined observation the robot has already contradicted, the robot executes them, and the executed actions feed both world models.

Figure 6 shows one such event under three protocols. We inject 1.2% uplink loss — a mild rate for a congested 5G uplink — and drop a single triggered frame. Without any protocol the divergence never recovers, crosses τ^* and the episode fails 1.8 s later; across 200 episodes at this loss rate, 34.2% end in a collision or timeout. Adding periodic forced resynchronization, which is the natural first fix and the analogue of an I-frame, bounds the excursion but does nothing about *detection*: the server spends an entire interval acting on a state it has no reason to distrust, and in 19.7% of episodes that interval contains a contact event. Sequence numbers, which cost two bytes, turn the same event into an 84 ms excursion. This is why §3.4 is a protocol contribution and not an implementation detail.

Takeaways. The deployed configuration needs 11 Mbps that commercial 5G supplies 23% of the time; the robot can afford a predictor 10^2 – $10^3\times$ cheaper than the policy, structurally so; action conditioning roughly doubles the usable window; fidelity ranks decision impact at $\rho_s=0.31$ and misclassifies 44% of frames in both directions; imagined observations are free for about 4 steps and catastrophic after 10; and unprotected silence turns 1.2% loss into 34.2% episode failures.

3 Design

ActDelta has four parts: a shared action-conditioned world model (§3.1), a decision-relevance trigger that replaces the fidelity test (§3.2), a bandwidth-adaptive threshold (§3.3), and the resynchronization protocol (§3.4). Figure 1 shows how they compose, and its numbered path is one control step.

- ① The robot encodes the true observation.
- ② Both ends roll the identical world model forward on the action chunks the policy has already emitted, so they stay aligned while the

uplink is silent. ③ A distilled head estimates how much the discrepancy between the two would move the policy’s action, and ④ only that estimate — never a reconstruction error — crosses the threshold. ⑤ When no frame arrives the server decodes $\hat{o}_t$ for the policy, and ⑥ the returned chunk drives the executor and, through it, both world models. The robot never runs the decoder and never runs the policy. The world model is not a contribution: action-conditioned predictors that generate policy-consumable observations from a rollout of committed actions already exist and match a policy’s input format [23, 40, 62]. We adopt one, shrink it, and claim nothing about it; our contributions are the trigger and the protocol, both indifferent to which predictor sits underneath.

3.1 The Shared Predictor

Both ends run identical weights W : an encoder E mapping three camera views to a 96-token latent z_t , a chunk-wise latent dynamics model $\hat{z}_t = W(\hat{z}_{t-1}, a_{t-1})$, and — on the server only — a decoder D that renders $\hat{o}_t$ in the format the policy expects. Total 68M parameters, 0.14 GB of weights, 11.7 ms per control step on the robot (Table 1). We chose the size to serve the argument of §2.2 rather than to maximize prediction quality: a smaller model makes the compute-asymmetry claim stronger, and neither the decoupling result nor the protocol depends on prediction quality. A large video diffusion model would predict better, fail to run on the robot, and invite the objection that the robot might as well run the policy.

The two ends agree because they share weights, an initial state and an action sequence: the actions travel down the link and the robot knows what it executed. Consistency in floating point does not follow from consistency in algorithm, and this bit us: the same graph on an Orin NX and on an A100 drifts apart by 4×10^{-4} per step in fp16. We quantize the recurrent latent to 8 bits after every step, which makes the rollout exactly reproducible across platforms for 0.4 pp of success.

3.2 Triggering on Decision Relevance

The quantity we want, and why we cannot have it. The ideal trigger fires when the true observation would change the policy’s action, i.e. on $\delta_t = \|\pi(o_t) - \pi(\hat{o}_t)\|$. Computing it requires two forward passes of the policy on the robot, which is exactly the thing the robot cannot do — this is not an efficiency concern but a definitional one, since a robot that could evaluate π twice per step would not be offloading. So δ_t is available offline, in quantity, and never online.

The distilled head. We therefore train a small head Φ to predict it. Given the encoded true observation z_t and the

predicted latent $\hat{z}_t$, both of which the robot already has,

$$\hat{\delta}_t = \Phi(z_t, \hat{z}_t, z_t - \hat{z}_t), \quad (2)$$

a 2.1M-parameter MLP over pooled latents, trained by regression on δ_t computed offline over LIBERO and DROID with the policy in the loop (1.4M frame pairs, 9 GPU-hours, once). At runtime it costs 1.1 ms and never touches the policy. A concurrent submission applies the same methodology — online-unavailable ground truth, obtainable proxy, offline calibration — to a different quantity in the same deployment [3]; the two are orthogonal, latency against bandwidth.

An obvious alternative is to compare in the policy's own visual-encoder latent space, which is decision-relevant by construction. We evaluate it and do not use it: a VLA's vision tower costs 214 GFLOP per step, 59× the rest of our pipeline, which would destroy the argument of §2.2. The head is a 200× cheaper approximation, and the gap is 1.6 pp.

The rule. The robot transmits when

$$\hat{\delta}_t > \tau(\hat{B}_t) \quad \text{or} \quad k_t \geq T_{\text{resync}} \quad \text{or} \quad \text{NACK received}, \quad (3)$$

where k_t counts steps since the last transmitted frame. The three clauses are the mechanism, the safety net, and the protocol respectively, and they are deliberately not merged: the first is an estimate and may be wrong, the second bounds how wrong it can be for how long, the third handles the link.

3.3 Making the Threshold a Network Citizen

A fixed τ would make ActDelta a source that ignores the link, which is the mistake the status quo already makes. We set

$$\tau(\hat{B}_t) = \text{clip}\left(\tau_0 \left(\frac{B_{\text{ref}}}{\hat{B}_t}\right)^\alpha, \tau_{\min}, \tau_{\max}\right), \quad (4)$$

with $\tau_0 = 0.15$, $\alpha = 0.5$, $[\tau_{\min}, \tau_{\max}] = [0.08, 0.30]$ and $\hat{B}_t$ an EWMA of delivered uplink throughput. When the link is scarce the threshold relaxes and the robot speaks only about frames that would definitely change the action; when the link is generous it tightens and the system approaches continuous streaming. The clip at $\tau_{\max}$ is what keeps the mechanism honest: the threshold is never allowed above 0.30, which is below the policy-flip level $\tau^* = 0.35$ of Figure 5(b), so degradation under congestion is bounded by construction rather than by tuning.

3.4 The Resynchronization Protocol

Figure 7 gives the state machine. Three mechanisms carry it. *Sequence numbers* on every transmitted frame let the server detect a gap and issue a NACK, which is the distinction §2.7 shows is missing from published event-triggered designs. *Heartbeats* every $T_h = 8$ control steps carry the current sequence number, the robot's own $\hat{\delta}_t$, and a rolling hash of

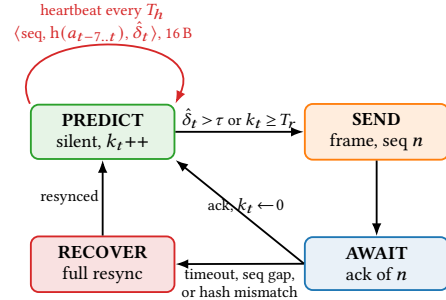


Figure 7: The protocol makes silence unambiguous. Heartbeats carry a sequence number and a rolling hash of the actions the robot actually executed, so the server can separate correct silence from loss and demand recovery.

the last eight executed actions; the hash catches the second divergence source, in which a lost or late *downlink* chunk makes the robot fall back to its safety behaviour so that the actions it executed are not the actions the server assumed. *Forced resynchronization* every $T_{\text{resync}} = 60$ steps (4 s) bounds drift in the case the trigger itself is wrong, which the head's finite accuracy guarantees will happen; it is the I-frame of this scheme, and its cost is a 1.7% floor on the send rate.

The resulting divergence is empirically bounded by

$$d_t \leq \tau(\hat{B}_t) + \varepsilon_\Phi + \beta \cdot \min(k_t, T_{\text{resync}}) \mathbf{1}[\Phi \text{ misses}], \quad (5)$$

where ε_Φ is the head's error (P99 = 0.09) and $\beta = 0.07/\text{step}$ the measured open-loop drift rate from Figure 3. The head misses on 1.3% of steps, so the third term is rare; the design rule is to pick T_{resync} so that the bound stays under τ^* at the offline-measured miss rate. §5.4 reports P99.9 divergence of 0.21 against a predicted bound of 0.24.

3.5 Composition, and What Breaks

ActDelta occupies the temporal axis — whether a frame is sent — and is orthogonal to the spatial axis, which decides what a sent frame costs. We run IT-RTC [8] underneath everything as a latency substrate, and we compose with a learned VLA-specific frame codec [31] on triggered frames; §5.2 evaluates both compositions and neither interferes.

Three failure modes are handled explicitly. On *cold start* the first two frames of every episode are transmitted unconditionally. If the *head is unavailable* or its *input stale*, the trigger degrades to $\hat{\delta}_t = \infty$ and the system falls back to continuous transmission — to the status quo, not to something worse. And if the *uplink cannot carry even the forced resynchronizations*, measured at sustained uplink below 0.6 Mbps, the robot stops rather than act on an unbounded-divergence plan; this is the convention production teleoperation stacks

use and it is enabled in every arm of our evaluation, baselines included.

4 Implementation

ActDelta is 5,200 lines of Python and 1,100 lines of C++, and reuses four public codebases rather than reimplementing them.

World model and policies. The predictor follows the chunk-wise action-conditioned architecture of policy-in-the-loop world models [40], reduced to 68M parameters (6 latent-transformer blocks, width 384, 96 tokens per step) and trained on the 2,000 LIBERO demonstrations plus a 40k-episode DROID subset for 19 hours on eight RTX 3090s. Policies are served through LeRobot [12], whose `policies.rtc` module provides the reference IT-RTC implementation for π_0 , $\pi_{0.5}$ and SmolVLA; OpenVLA-7B is wrapped behind the same interface. The server is organized after the unified policy-server design of VLAagents [27], one process per GPU with a shared request queue, which is also our status-quo baseline.

Robot client. The encoder, rollout, head and trigger run in the control thread on a Jetson Orin NX as a single TensorRT engine with the 8-bit latent quantization of §3.1 applied between steps; this is the C++ part, because it is the only code with a real-time deadline. The client holds no policy weights.

Transport. The uplink is a userspace UDP path with a 16-byte control header carrying sequence number, action hash and $\hat{\delta}_t$, and a JPEG or learned-codec payload for triggered frames. NACKs and forced-resync requests travel on the downlink alongside action chunks. Trace replay uses mahimahi [48] with per-direction link shells driven by the Irish, Lumos5G and MN-Wild uplink series; loss and delay injection uses `tc netem`. We deliberately do not run over TCP, unlike the status quo [27]: head-of-line blocking on a link this scarce turns one lost frame into a multi-hundred-millisecond stall, and our protocol already provides the loss detection that TCP would be supplying.

Compute-budget emulation. The ρ sweep needs robot budgets we do not own hardware for. We emulate them by pinning the client to a fraction of the Orin's SMs via MPS and capping the clock, calibrating each setting against measured FLOP/s so that the reported ρ is achieved rather than nominal. The $\rho = 0.109$ point is native hardware and anchors the curve.

5 Evaluation

We ask five questions: does a decision-relevance trigger beat a fidelity trigger when nothing else differs (§5.1); where does ActDelta sit against the full range of alternatives (§5.2); over

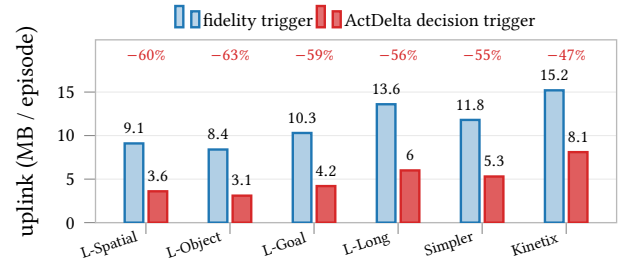


Figure 8: The one comparison that isolates the paper's mechanism. Identical world model, policy, link and protocol; each arm's threshold is tuned so that the two match on task success within 0.7 pp.

what compute ratios does the mechanism pay (§5.3); does the protocol do what §3.4 claims (§5.4); and what does it cost, in simulation and on hardware (§5.5–§5.7)?

Setup and baselines. Environments, policies, traces and hardware are as in §2.3; every configuration runs at least 50 episodes, error bars are 95% bootstrap CIs, and IT-RTC is enabled everywhere. An episode is 50 s, 750 control steps, and a transmitted frame is 137 kB unless a codec says otherwise. The baselines fall into three groups, and the grouping is the argument. *Status quo and conventional coding*: (1) the deployed policy server, fixed 10 Hz JPEG over TCP [27]; (2) fixed 10 Hz H.265 with rate control [57]; (3) pixel-difference change detection. *Classical predictive transmission*: (4) send-on-delta with a Taylor predictor on the proprioceptive state [43, 56]; (5) the same rule lifted to the image latent. *Modern generative and spatial competitors*: (6) a world-model transmitter with a fidelity trigger — our reproduction of [26] on the identical world model as ours; (7) learned image compression for VLA [31]; (8) ActDelta; (9) ActDelta composed with (7).

5.1 Fidelity vs. Decision: The Single-Variable Result

We put this first because it is the paper's only clean evidence for the mechanism; every other comparison confounds it with something else. Figure 8 holds the world model, the policy, the link, the protocol and the success rate fixed, and varies only the quantity that crosses the threshold. Averaged over the four LIBERO suites, triggering on estimated decision impact rather than on reconstruction error sends 59.2% fewer uplink bytes at the same task success.

The number is the direct consequence of Figure 4: the fidelity trigger transmits the 31.7% of frames that are visually wrong and behaviourally irrelevant, and ActDelta does not. Worth noting is what *did not* happen — success did not fall. A trigger that merely sent less would have paid for it; ActDelta does not, because it spends part of the saving on the 12.4% of

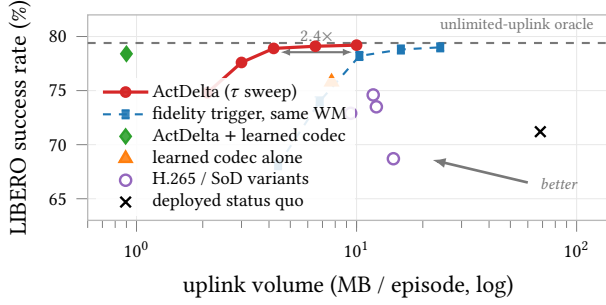


Figure 9: The byte–success frontier. ActDelta dominates every baseline at every operating point; each curve is a sweep of that arm’s own threshold.

frames a fidelity trigger skips, so its send rate (4.1% of control steps) is a differently-shaped set rather than a scaled-down version of the fidelity trigger’s (10.0%). Kinetix is the honest boundary case: its dynamics shorten the usable window to 2.6 steps, both arms transmit far more often, and the advantage falls to 46.7%. We expect any domain with sub-200 ms scene dynamics to behave this way.

5.2 The Full Frontier

Table 2 and Figure 9 place that mechanism in context. Both report π_0 with IT-RTC over the Irish 5G traces; uplink is MB per episode and the LIBERO column averages the four suites. Four observations. First, the classical baselines behave exactly as their assumptions predict: send-on-delta on the proprioceptive state (4) is the worst arm in the table, because a threshold on joint positions is blind to everything the cameras exist to see, and lifting the same rule to the image latent (5) recovers most of the loss but inherits the fidelity criterion and with it the misallocation of §2.5. Second, conventional video coding (2) is stronger than one might expect — H.265 exploits the same temporal redundancy with no model at all — and remains 2.8× more expensive than ActDelta because it still pays for every frame. Third, the spatial competitor (7) is genuinely good and genuinely orthogonal, and the combination (9) reaches 0.9 MB per episode, 75× below the status quo, at a 0.5 pp success cost we attribute to codec artifacts on the frames that matter most.

Fourth, and least expected: ActDelta *improves* on the status quo’s success rate by 7.7 pp while sending 16× less. The status quo is not merely wasteful but self-harming — at 11 Mbps demanded from a link that supplies it 23% of the time, 41% of its uploads arrive stale (§2.1). Sending less, when what you send is chosen well, makes it arrive on time.

The frontier adds what the table cannot show: across the whole sweep the two world-model arms stay separated by a roughly constant factor of 2.4 in bytes, the signature of a

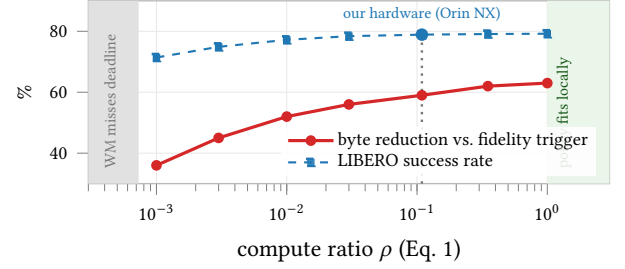


Figure 10: Benefit versus compute ratio, with both failure boundaries. Shading marks the region in which the world model cannot finish inside the control period and ActDelta degrades to fixed-rate transmission.

criterion difference rather than of a tuning difference — no threshold setting moves the fidelity arm onto our curve. We operate at the 4.2 MB knee; pushing τ below 0.15 buys 0.3 pp for 55% more bytes.

5.3 Where the Mechanism Applies

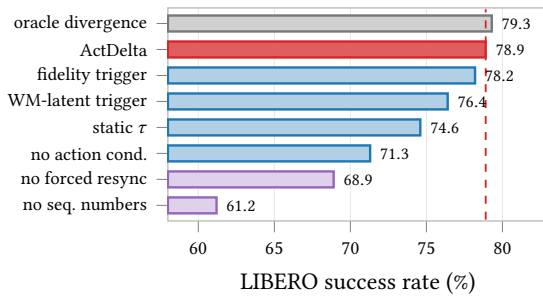
Figure 10 answers the question this system model always provokes. The benefit is not a knife-edge property of one hardware configuration: it stays above 50% across two and a half decades of ρ and saturates rather than peaks, because past a point the world model is good enough and more compute buys prediction quality the decision trigger does not need. The rise is steepest up to $\rho \approx 10^{-2}$, the point at which a model large enough for a 5-step window fits inside the budget. The lower boundary is sharp: at $\rho < \rho_{\min} = 7.3 \times 10^{-4}$ the predictor misses the 66.7 ms control deadline and the right engineering answer is a fixed-rate encoder, not this paper. The upper boundary is soft and is not where the value goes to zero — at $\rho = 1$ the byte reduction is still 63%, and only the reason to want it changes (§7). What the sweep says, in one sentence, is that the mechanism is scale-relative: what matters is that the policy is larger than the robot’s budget, not how large either one is.

5.4 Does the Protocol Earn Its Complexity?

We sweep uplink loss from 0 to 5% with tc_netem at a fixed trigger. Without sequence numbers or forced resynchronization — the configuration published event-triggered designs describe — the episode catastrophe rate (collision or timeout) rises from 1.8% at zero loss to 34.2% at 1.2% loss and 71.5% at 5%. Forced resynchronization alone cuts these to 9.4% and 22.1%: it bounds the damage but cannot detect it, so the server acts on a diverged state for up to a full interval. The full protocol reaches 2.1% and 4.6% with a median detect-to-recover time of 84 ms, for 240 B/s of heartbeat traffic — 0.02% of what it protects.

Table 2: End-to-end results. Send rate is the % of control steps at which a frame is transmitted; divergence is measured in the decision space of §3.2, where 0.35 flips the action.

	Transmission policy	LIBERO	SimplerEnv	Kinetix	Uplink	Send rate	P99.9 div.	vs. status quo
	Unlimited-uplink oracle (upper bound)	79.4	70.6	68.2	—	100%	—	—
status quo	(1) Policy server, 10 Hz JPEG/TCP [27]	71.2	62.8	55.1	68.4	66.7%	—	1×
	(2) Fixed 10 Hz H.265, rate-controlled	74.6	65.9	58.7	11.9	66.7%	—	5.7×
	(3) Pixel-difference change detection	72.9	64.1	57.0	9.4	9.1%	0.61	7.3×
classic	(4) Send-on-delta + Taylor, proprioceptive [56]	68.7	60.3	52.4	14.7	14.3%	0.74	4.7×
	(5) Send-on-delta lifted to image latent	73.5	65.0	58.1	12.3	12.0%	0.44	5.6×
modern	(6) World model + fidelity trigger [26]	78.2	69.1	63.4	10.3	10.0%	0.29	6.6×
	(7) Learned image compression for VLA [31]	75.8	67.2	60.8	7.7	66.7%	—	8.9×
	(8) ActDelta	78.9	69.8	64.6	4.2	4.1%	0.21	16.3×
	(9) ActDelta + (7)	78.4	69.4	64.0	0.9	4.1%	0.21	75.2×

**Figure 11: Ablation at 1.2% uplink loss, so that the protocol components are exercised. Purple marks the two protocol rows, which lose success *without* saving bytes.**

The divergence bound of Equation 5 also holds: over 2,400 episodes the P99.9 decision-space divergence is 0.21 against a predicted bound of 0.24 and a policy-flip threshold of 0.35, and the maximum observed value, 0.33, occurred in the one configuration where the head’s miss rate exceeded its offline estimate. Sensitivity to T_{resync} is mild between 30 and 120 steps (± 0.9 pp success, $\pm 1.4\%$ bytes) and degrades sharply past 180 steps, where Equation 5 predicts the bound crossing τ^* and success duly falls by 6.1 pp.

5.5 Ablations

Figure 11 separates the components, and the ordering is informative. The largest single loss comes from removing sequence numbers (-17.7 pp), which is not a performance knob but the difference between a protocol that can detect divergence and one that cannot; removing action conditioning costs 7.6 pp and 64% more bytes, confirming Figure 3. Replacing the distilled head with the world model’s own latent distance — the most natural way to build this system without our measurement — costs 2.5 pp and 69% more bytes, and replacing it with a fidelity criterion costs 145% more bytes, the same 59% reduction seen from the other direction. A static τ costs 4.3 pp and 38% more bytes.

Table 3: Robot-side cost per control step, Orin NX, median of 10^5 invocations; the 15 Hz control period is 66.7 ms.

Component	Latency	Memory	Energy
Encoder E , three views	6.9 ms	0.11 GB	0.11 J
Latent rollout W , one step	3.4 ms	0.06 GB	0.05 J
Decision head Φ + trigger	1.1 ms	0.02 GB	0.02 J
Protocol (seq, hash, framing)	0.3 ms	<0.01 GB	0.01 J
Total	11.7 ms	0.21 GB	0.19 J
as % of control period	17.5%	—	—

Two rows must be read with their uplink volumes, which the figure does not draw. The protocol ablations send 4.0 and 3.4 MB against the full system’s 4.2, so their 17.7 and 10.0 pp losses buy nothing — a correctness bug rather than a trade-off, and the reason we treat the protocol as part of the mechanism. The oracle row, which substitutes the ground-truth divergence of §2.5 for the head, bounds what a better estimator could add at 0.4 pp and 10% of bytes: the head is close enough to its own ceiling that the headroom is elsewhere.

Three further sweeps. *World-model size*: at 18M the prediction window shrinks to 3.1 steps and success falls to 76.1%; at 210M it reaches 79.2% but misses the control period on the Orin — precisely the trade ρ describes. *Latent quantization*: the 8-bit state that buys exact cross-platform reproducibility costs 0.4 pp, and without it 3.1% of episodes accumulate silent numerical divergence. *Head transfer*: a head distilled on π_0 and applied unchanged to OpenVLA-7B costs 2.3 pp and 8% more bytes, and one trained jointly on three policies recovers all but 0.6 pp: the calibration is policy-sensitive, not policy-specific.

5.6 Cost on the Robot

Table 3 reports the price of running a predictor on the robot, in full, because the honest accounting is less flattering than

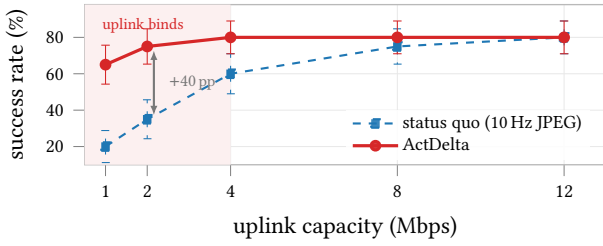


Figure 12: Franka Panda, 160 trials. Two tasks against five shaped uplink capacities; bars are 95% CIs.

the byte numbers. ActDelta costs 17.5% of the control period and 2.85 W of sustained compute. It saves radio energy — the status quo’s 10 uploads per second cost 5.2 W of 5G transmission against 0.32 W for ours — so the net is 5.5 W down to 3.2 W, a 1.7 $\times$ reduction rather than the 16 $\times$ the byte numbers suggest. Evaluating the trigger at 7.5 Hz halves the compute to 1.42 W for 0.4 pp of success, bringing the net to 2.6 $\times$. ActDelta trades robot compute for uplink capacity, and on a shared cell the uplink is the scarce resource, not the robot’s watts.

5.7 Physical-Robot Evaluation

Simulated contact dynamics are only as trustworthy as the simulator, so we ran a controlled hardware experiment on a Franka Panda — the DROID platform, chosen so that the offline analysis and the hardware share an embodiment — with two tasks (peg insertion and cluttered pick-and-place), five shaped uplink capacities and 160 trials. This is a variable sweep, not a data collection: nothing is trained and no dataset is built. The result we care about in Figure 12 is not the peak but the shape. The two arms converge above 8 Mbps, as they must; below 4 Mbps the status quo degrades steeply while ActDelta stays nearly flat, because its send rate is set by what the scene does rather than by a clock. At 2 Mbps — a congested commercial uplink — the status quo succeeds on 35% of trials and ActDelta on 75%, sending 0.44 MB per episode against 5.8 MB. That measured byte ratio, 13.2 $\times$, is within 20% of simulation’s 16.3 $\times$.

6 Related Work

Send-on-delta and event-triggered sampling. We did not invent this mechanism class. Lebesgue sampling [4, 5] and Miskowicz’s send-on-delta [43] established transmitting on signal deviation rather than on a clock; predictive variants transmit when the signal departs from an extrapolation the receiver can reproduce, with truncated Taylor [56] or linear [49] predictors, and derive thresholds from error bounds [44]; controlled-communication policies solve for an optimal transmission policy against a local copy of the

remote estimate [25, 41, 45, 58]. Every element of that setting differs from ours in the same direction: the payload is a low-dimensional state, not multi-camera video; the predictor is analytic, not learned; the receiver is an estimator with an error covariance, not a black-box 7B policy; and the trigger comes from an estimation bound rather than from a decision. We take this line as baseline group (4)–(5) rather than as background, and inherit its open problem: that event-based sampling cannot detect dropouts for want of a shared periodic timestamp [50] is what §3.4 addresses.

World models and generative semantic communication. The nearest neighbour is world-model-aided semantic transmission, which runs a generative world model at both ends, omits transmission while prediction stays reliable, and adds a feedback module to decide when the current frame must be sent [26], and a broader literature transmits sketches, keyframes or tokens and reconstructs generatively [9, 15, 21, 33, 34, 39, 64, 65]. What unites them is a *passive* receiver: the reconstruction is rendered, scored by PSNR or perceptual or semantic consistency, and does not act. Ours acts, and its actions determine the next observation, which is what makes reconstruction fidelity the wrong trigger (§2.5) rather than an imperfect one. We reproduce [26]’s design as baseline (6), with our world model, so that the comparison isolates the criterion.

The exception is a world-model digital twin that selects semantic tokens by their marginal contribution to expected long-term return, normalized per bit [61]: the closest anyone comes to our criterion, and explicit that a one-shot fidelity metric is the wrong one for a closed loop. It differs in what it estimates and in what it decides — a learned long-horizon return, from an actor–critic it trains inside the twin, against our one-step divergence in the action space of a policy we never retrain; and *which tokens* a transmission carries, against *whether to transmit at all*, the decision that makes silence ambiguous and forces §3.4.

Predictive display and model-mediated teleoperation. Predicting the remote scene to hide delay is the oldest line here, from the phantom robot [6] and passivity-based teleoperation [2, 24] through environment-adapted controllers [11, 51] to recent zero-shot benchmarks of video models as predictive displays [29]. The prediction there is for a human, the purpose is to compensate perceived latency, and the evaluation is operator performance and cognitive load. We predict for a model, to avoid sending bytes, which makes our criterion automatable and calibratable in a way theirs cannot be.

Observation compression for VLA.. A fast-growing line compresses what an offloaded policy consumes. Learned image compression tuned for VLA notes that general-purpose

codecs optimize perceptual fidelity rather than downstream control and that visual importance is highly non-uniform across cameras and regions [31]; token-level methods distill each frame to a single latent token [60], cache or prune visual tokens [19, 63], or compress task-oriented features for device–edge and vehicle-to-everything collaboration [66, 69]. All allocate bits *within* a frame; none decides whether the frame is worth sending. The axes are orthogonal and Table 2 row (9) shows they multiply. The same holds for video-analytics systems that filter on-camera or let the server drive region selection [14, 18, 37, 67]: their consumer scores per-frame accuracy, not a closed loop in which its next input is a consequence of its own output.

Action-conditioned world models. Policy-in-the-loop world models generate futures chunk by chunk from a policy’s own action rollout and match the policy’s input format [40], building on a line of learned simulators [10, 22, 23, 62]. We use one as a component: our results are stated relative to a fixed predictor and would move with a better one without changing sign, the property §2.5 was designed to guarantee.

Edge serving and latency tolerance for VLA.. Splitting inference between device and edge is long established [28]; recent systems specialize it for policies with latency-tolerant cloud–edge collaboration [52], policy servers [27], KV-cache management [36], pipelined edge inference [17] and deadline-aware scheduling [13], all under the deployment constraint a recent survey takes as given [20]. A parallel line makes the policy tolerate delay rather than reduce it [1, 8, 59]; we run IT-RTC underneath every arm because it is a substrate, not a competitor. Closest in method is a concurrent submission that reads a value-of-information signal off a policy’s own action chunks to decide *when* an inference is worth requesting [3]: same calibration methodology, latency axis instead of bandwidth.

7 Discussion and Limitations

The upper bound is the world model. Everything ActDelta saves is bounded by how long the shared predictor stays useful, and Figure 8’s Kinetix column is the honest version of this: when scene dynamics outrun the prediction window, both triggers converge toward continuous transmission and our advantage shrinks from 59% to 47%. We expect the mechanism to suit fast locomotion and scenes with independently moving agents poorly — precisely the settings where the action conditioning of Figure 3 stops paying.

At $\rho \geq 1$ the mechanism survives; the motivation changes. Figure 10 shows the byte reduction holding at 63% when the robot could run the policy locally, and it is fair to ask why anyone would then offload. Compute is not the

only reason to centralize: pushing 15 GB of weights to every robot for every version is an operational cost a task-agnostic 0.14 GB world model does not incur; frontier weights are often not distributed at all; centralized execution is what lets fleet experience be aggregated; and one server amortized over N robots is the deployment in which uplink is genuinely contended. We did not evaluate the multi-robot case and consider it the most interesting missing experiment.

What the head is, and is not. Φ estimates a quantity defined by a specific policy, so it inherits that policy’s blind spots: if the policy is indifferent to something it should attend to, ActDelta will happily not transmit it. The head is also trained, which sits uneasily beside our claim that the world model’s quality does not matter — it does not matter for Figure 4, which is measurement, but it does matter for the system. A head miscalibrated for a new embodiment degrades toward the world-model-latent trigger, which §5.5 measures at -2.5 pp rather than at collapse, so the failure is graceful; we would still want an online confidence check before deploying on hardware we have not calibrated.

What we did not evaluate. A compromised client can suppress transmission by reporting low $\hat{\delta}_t$; server-side plausibility checks against the heartbeat’s own divergence estimate would catch it and are unimplemented. Every closed-loop result is a 7-DoF arm. And our physical experiment shapes uplink capacity rather than replaying a live cellular link, so it validates the response to scarcity and not the interaction with real radio dynamics.

8 Conclusion

Predictive transmission has been asking the same question for twenty years: how wrong is the receiver’s reconstruction? That question is right when the receiver renders what it receives, and wrong when the receiver is a policy that acts on what it receives and thereby chooses what it receives next. We measured the difference — fidelity ranks decision impact at $\rho_s = 0.31$ and misclassifies 44% of frames in both directions — and built the transmitter that asks the other question: a shared action-conditioned world model, a distilled robot-side estimate of how much a frame would move the policy’s action, a bandwidth-adaptive threshold, and the resynchronization protocol that closed-loop prediction needs and event-triggered sampling never had. At the same world model and the same task success, that change of criterion alone is worth 59% of the uplink, $16\times$ against the deployed status quo and $75\times$ composed with spatial compression. The general claim is shorter: when the consumer of a stream is a decision-maker, the transmitter should be measuring decisions, not pixels — and it can, cheaply, without ever running the decision-maker.

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